One photo in, a full focal stack out

Goal: click any pixel and bring it into focus

Idea: a focal sweep is a 9-frame “video”, so fine-tune a video diffusion model (Stable Video Diffusion)

Trained: only the temporal blocks; image encoder and decoder stay frozen



pos 0 · near pos 4 pos 8 · far

Three of the nine frames in one real training stack



Based on Tedla, Zhang, Zhang & Xin, “Learning to Refocus with Video Diffusion Models,” SIGGRAPH Asia 2025 · learn2refocus.github.io

Slot conditioning beats motion-bucket encoding

We fine-tuned three ways of telling the model the input's focus position, then tested on 20 held-out scenes × 3 input positions

Model	PSNR ↑	SSIM ↑	LPIPS ↓	Depth ρ
Authors' 200k checkpoint	31.07	0.893	0.160	−0.70
A random slot (ours)	28.76	**0.870**	**0.192**	−0.61
B position ablated (ours)	**29.11**	0.863	0.196	−0.58
C motion_bucket_id (ours)	28.62	0.858	0.199	−0.61

Bold = best of our three arms. Depth ρ: how well the sharpest frame follows scene depth (real stacks: −0.76; more negative is better).

- B beats C on every metric (+0.49 dB, 55 of 60 items); A beats C on SSIM and LPIPS.
- A vs B is a trade-off: B has higher PSNR (+0.35 dB), A better SSIM and LPIPS.
- The authors' model leads everywhere; ours trained 2% as long.

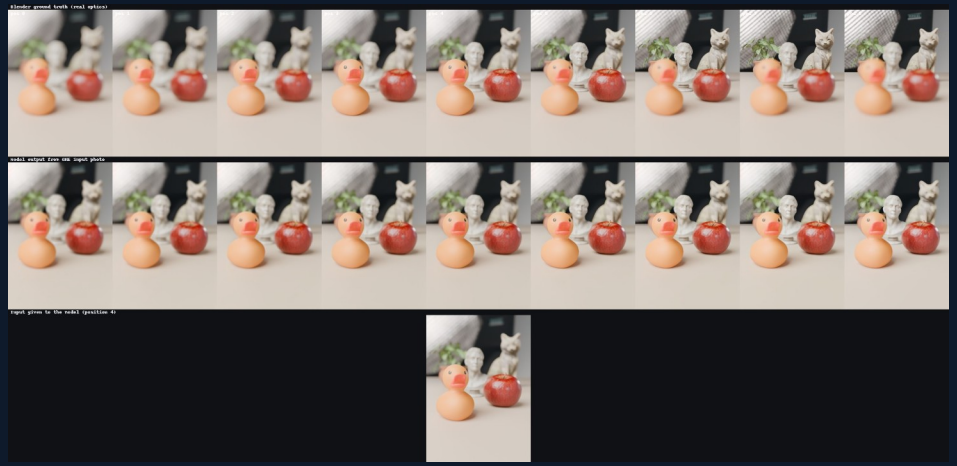
8 × A100
3.4 h · about $54

4,000
steps per arm (2% of the 200k)

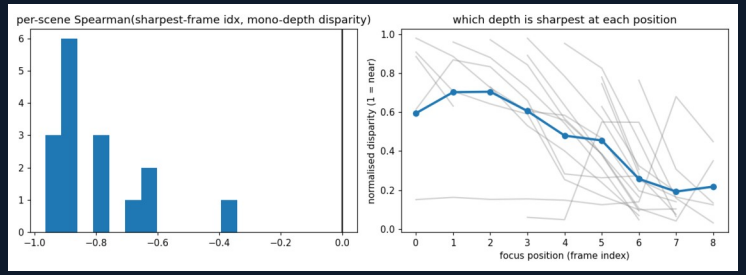
20 × 3
scenes × input positions

Based on Tedla, Zhang, Zhang & Xin, "Learning to Refocus with Video Diffusion Models," SIGGRAPH Asia 2025 · learn2refocus.github.io

Demo, a depth check, and the caveats



Blender scene: true focal stack (top) vs. model output from one photo



Extra credit: the sharpest frame tracks monocular depth (Depth Anything V2), Spearman ρ = −0.80 on 16 scenes

Click-to-refocus viewer
Click a pixel; the page shows the nearest of the 9 frames.

Synthetic test
Blender scenes with a real sweep: plausible near-to-far order, peaks off by 1–3 positions.

Limits
Short training, one seed, 20 scenes; Blender distances are our assumption.

Based on Tedla, Zhang, Zhang & Xin, “Learning to Refocus with Video Diffusion Models,” SIGGRAPH Asia 2025 · learn2refocus.github.io